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An Enhanced Approach to Address Cementing Challenges While Ensuring Well Integrity Mandates for SWAG and CCS Projects by Utilizing a Combination of Cement and Resin, Implemented for the First Time in Field Operator in the UAE

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Abstract

Drilling and completing the wells for CCS and SWAG poses a unique challenge of extreme exposure to higher CO₂ wherein utilization of conventional Portland based cement systems can compromise of well integrity and well life span. For such complex applications an innovative approach by combining cement and resin to address the carbonation challenges was experimented for the First Time in Field Operator in the UAE.

Carbonation in cement can increase permeability and decrease compressive strength, leading to potential structural vulnerabilities. A new composite of resin and reduced Portland cement was designed specifically to resist carbonation and enhance elasticity and shear bond strength. This innovative blend aims to address common issues related to cement integrity in turn enhancement of well life cycle. Extensive tests were conducted which includes phenolphthalein analysis to assess pH changes, mechanical properties examination, permeability and porosity assessments, as well as long-term CO₂ exposure testing to evaluate the composite's resilience under harsh environmental conditions.

Due to increasing regulatory pressure to reduce carbon dioxide emissions while maintaining integrity, operators have shifted their focus to Carbon Capture and Storage (CCS) projects. Additionally, SWAG wells are required for enhanced oil recovery, which has become essential for operating companies. Both categories of wells (CCS and SWAG) face high levels of CO₂ exposure, necessitating innovative approaches to meet cementing requirements under such harsh conditions. Proposed cement systems have proven to be a game-changing solution.

Utilizing CO₂-resistant slurry for the cementation of 9-5/8" casing and 7" liner marks the first use of this technology in the UAE region. This approach aims to reduce the impact of carbonation on the cement structure and promote sustainability. Computer-aided pressure, temperature, and placement modeling were

performed to achieve a fit-for-purpose design. The inclusion of resin in the slurry minimizes the carbonation of the cement slurry while enhancing its mechanical properties.

Through optimization of cement slurry properties, rigorous lab testing, and excellent operational planning, the engineered solutions resulted in successful cement placement. This was validated by post job cement integrity evaluation that showed excellent bonding and isolation across all reservoirs, ensuring robust cement sheath integrity.

Introduction

Drilling and completing the wells for CCS and SWAG poses a unique challenge of extreme exposure to higher CO₂ wherein utilization of conventional Portland based cement systems can compromise of well integrity and well life span. For such complex applications an innovative approach by combining cement and resin to address the carbonation challenges was experimented for the First Time in Field Operator Field Operator in the UAE.

SWAG Wells

Simultaneous Water Alternating Gas (SWAG) injection wells are designed to optimize hydrocarbon recovery and improve reservoir management by alternating the injection of water and gas. This approach is particularly effective in maintaining reservoir pressure and enhancing sweep efficiency, ensuring that more hydrocarbons can be efficiently extracted from subsurface formations. However, the construction and completion of SWAG wells introduce a distinct set of technical challenges, especially in environments where high concentrations of CO₂ are present.

In recent years, operators have faced the unique challenge of drilling and completing wells destined for Carbon Capture and Storage (CCS) and SWAG operations within fields characterized by elevated levels of CO₂. In such environments, the long-term integrity of conventional Portland cement-based systems is often compromised, as these materials are susceptible to chemical degradation and carbonation over time, leading to potential wellbore integrity issues and a reduction in the productive lifespan of the well.

To address these challenges, operators engineering teams explored innovative material solutions, building upon traditional cementing practices. For the first time in the operator's onshore fields, a hybrid approach combining both cement and resin was trialed. This novel method was specifically engineered to counteract the aggressive effects of carbon dioxide on wellbore cement, thereby significantly enhancing resistance to carbonation and extending the operational life of the well. The resin component, when blended with cement, provides added chemical resilience and improved mechanical properties, ensuring a durable seal against fluid migration and casing corrosion in high-CO₂ environments. The implementation process included meticulous laboratory testing, material compatibility studies, and field trials to validate performance under anticipated downhole conditions.

This innovative cement-resin blend was carefully pumped into the annulus during primary cementing operations, establishing a robust barrier that not only anchors the casing string but also withstands the harsh geochemical environment encountered in CCS and SWAG wells. The successful deployment of this system marks a pivotal advancement in well construction technology, setting a precedent for future developments in the region and providing a blueprint for other operators facing similar challenges worldwide.

This pioneering approach is a testament to the company's commitment to operational excellence, environmental stewardship, and the pursuit of sustainable energy solutions. It also demonstrates the importance of interdisciplinary collaboration among drilling engineers, material scientists, and field operations teams in delivering fit-for-purpose solutions tailored to the demanding conditions of the UAE's onshore assets.

The company is committed to innovating solutions for reducing greenhouse gas emissions and enhancing sustainability, all while capitalizing on oil production. This dual approach ensures that while we continue

to extract valuable resources, we do so with a keen focus on efficiency and environmental responsibility. The project encompasses the drilling of four wells, comprising one water injector, one CO₂ injector, and two oil producers.

In parallel, we are planning to drill four observer wells to monitor the behavior of the main four wells. This approach ensures a comprehensive understanding of the subsurface interactions and the efficiency of the injection and production processes. The Water and CO₂ injectors are planned to be drilled within the same reservoir subunit, with the water injector strategically placed above the CO₂ injector to maintain the CO₂ beneath the water layer, preventing it from dispersing into unintended reservoirs. The primary goal of this project is to ensure the CO₂ remains effectively contained, optimizing the injection process and enhancing overall reservoir performance.

The project involves drilling a total of four primary wells: one water injector, one CO₂ injector, and two oil production wells. Additionally, four observation wells will be drilled to monitor the performance and interactions of the main wells. Both the water and CO₂ injectors are located within the same reservoir subunit, with the water injector positioned above the CO₂ injector. This strategic placement ensures that the injected CO₂ remains contained beneath the water layer, reducing the risk of it migrating into unintended reservoir zones. The overarching objective is to optimize the injection processes, maximize hydrocarbon recovery, and maintain effective containment of CO₂, thus improving reservoir performance and minimizing environmental impacts.

By integrating advanced drilling techniques and innovative well designs, the project aims to overcome the technical challenges posed by high-CO₂ environments. The use of a hybrid cement-resin system provides enhanced protection against carbonation and casing corrosion, leading to improved well integrity and a longer operational life. Comprehensive laboratory testing, compatibility studies, and field trials have validated this solution under expected downhole conditions, ensuring its reliability.

Furthermore, the company's commitment to sustainability is demonstrated through efforts to reduce greenhouse gas emissions while maintaining high standards of oil production. This dual focus on operational efficiency and environmental responsibility positions operators as a leader in advancing sustainable energy practices in the UAE and sets a benchmark for similar projects globally.

Carbon Dioxide (CO₂) Emissions and Cement Integrity in CCUS Wells

CO₂ emissions are a major contributor to climate change. One effective mitigation strategy is Carbon Capture, Utilisation, and Storage (CCUS), which involves injecting CO₂ into geological formations such as deep saline aquifers, un-mineable coal seams, and depleted oil/gas reservoirs.

However, Portland cement, commonly used in CO₂ injection wells, is highly vulnerable to acidic attack. CO₂ can penetrate the cement matrix, and when dissolved in pore water, forms carbonic acid (H₂CO₃). This acid reacts with calcium hydroxide (Ca(OH)₂) in the cement, initially forming calcium carbonate (CaCO₃), which can further convert to calcium bicarbonate (Ca(HCO₃)₂). The latter is highly soluble and leaches out, increasing porosity and permeability, and ultimately compromising mechanical strength and structural integrity.

To address this, polymer-based admixtures can be added to cement paste. These polymers enhance both mechanical and chemical resistance, forming a monolithic matrix where the polymer phase interpenetrates the cement hydrates. Studies by Knapen et al. show that even low concentrations of water-insoluble polymers (<4%) can form polymer bridges between Ca(OH)₂ crystals, improving cohesion and reducing cracking.

Epoxy resins are particularly promising due to their high strength, bonding capability, and chemical resistance. Traditionally, they require a hardener (amines or organic acids) to cure. However, research indicates that epoxy resins can also cure in the presence of alkalis like Ca(OH)₂, enabling hardener-free

formulations. These systems offer self-healing properties, where unreacted resin in the pores activates upon crack formation, reacting with $\text{Ca}(\text{OH})_2$ to seal the cracks.

Hardener-free epoxy-modified cements have demonstrated superior durability. For example, at a 40% polymer- cement ratio, the carbonation depth was found to be 1 mm without hardener, compared to 3 mm with hardener, after one year of exposure.

Most carbonation studies use mild conditions (20% CO_2 , 20°C, atmospheric pressure), often with polymers as surface coatings. However, few have examined polymer admixtures under realistic disposal conditions—high temperature and pressure. To fill this gap, an experimental study was conducted using epoxy resin blends (with and without hardener) in cement paste, exposed to CO_2 -saturated water at 70°C and 50 bar. Results were compared against unmodified cement paste to assess degradation resistance.

S1 Composites - Without Hardener

- These are epoxy resin-cement mixtures without any added hardener.
- The epoxy resin may partially cure due to interaction with cement hydration products (like $\text{Ca}(\text{OH})_2$), but not fully.
- Performance: These composites showed higher degradation when exposed to CO_2 . The resin tended to leach out, increasing porosity and allowing deeper CO_2 penetration.
- Degraded layer: Often thicker than pure Portland cement, especially at higher resin contents (e.g., 30% resin led to ~2385 μm degradation).

S2 Composites - With Hardener

- These mixtures include epoxy resin plus a hardener, which initiates full curing and crosslinking of the resin.
- The result is a stronger, more chemically resistant matrix.
- Performance: These composites had lower degradation under CO_2 exposure. The cured resin formed polymer bridges and reduced porosity.
- Degraded layer: Smaller than pure Portland cement (e.g., ~665 μm at 20-30% resin content).
- For Cementing in CO_2 Wells, S2 composites are more suitable for CO_2 -rich environments due to their enhanced durability. S1 composites may offer self-healing potential but are less reliable under high- pressure CO_2 conditions.

Table 1—Cementing Decision Matrix for CO_2 Injection Wells

Criteria	Portland Cement	Epoxy- Modified Cement (S1)	Epoxy-Modified Cement (S2)	High-Alumina Cement	Modified Portland (Latex, Silica, Diluents)
CO_2 Resistance	Moderate (carbonation risk)	Poor (resin leaches, high degradation)	Good (crosslinked resin, low degradation)	Excellent (no Ca to react with CO_2)	Good (low permeability, partial carbonation)
Availability	High	Moderate	Moderate	Low (special order)	High
Additive Compatibility	High	Low	Low	Low	High
Self-Healing Potential	None	Yes (uncured resin may heal cracks)	Limited	None	Possible with swelling additives
Field Proven	Yes (SACROC study)	Lab only	Lab only	Field tested (SPE-91861)	Field tested (SPE- 119267)

Criteria	Portland Cement	Epoxy- Modified Cement (S1)	Epoxy-Modified Cement (S2)	High-Alumina Cement	Modified Portland (Latex, Silica, Diluents)
Operational Complexity	Low	Moderate	High (precise mixing)	High (QC, logistics)	Moderate
Recommended Resin Content	N/A	<20%	<30%	N/A	N/A
Interface Integrity	Critical	Weak	Strong	Strong	Strong

Well Engineering and Construction

As CCS/SWAG well construction increases, material selection and well design are critical to deliver an annular barrier that is robust enough to withstand decades of injection cycles and can meet long-term storage requirements. The first objective for CCS well construction barrier design is to minimize the opportunity for chemical corrosion.

Challenges

- SWAG wells drilled by operator will be subjected to exposure of CO₂ with concentrations up to 99.9%
- No standard API procedure is available for testing the CO₂ Resistant slurries.
- First time in UAE slurries mixed with Resin+ Cement to be designed and executed.
- Achieving the fit to purpose design due to challenges in mixing and specialized mixing procedures for the job.
- Severe losses while drilling 8 1/2 section.

Cement job Preparation and planning

Using the cement decision matrix plan was made to utilise cement plus resin system (with hardener) for these type of applications. To reduce the risk of carbonation and alteration of the sheath, the cement composition was tailored using specific design. These included modification of the hydration products of Portland cement, minimization of permeability and inclusion of epoxy-amine resin additives for microstructural modification of the hydrated cement sheath.

The primary hydration products of Portland cement are calcium silicate hydrate (CSH), calcium hydroxide (CH), ettringite and monosulfate. For an API class G or H cement, CSH represents about 50 to 60% of the hydrated solids volume, with CH comprising 20 to 25% of the hydrated solids volume. The remainder is primarily ettringite and monosulfate.

Of the hydration products CSH has the highest resistance to degradation by carbonic acid exposure but may decalcify over long periods of exposure forming silica gel and releasing calcium ions. CH undergoes the fastest reaction with carbonic acid to form calcium carbonate, which may serve to temporarily densify strengthen the cement matrix, but further reaction with carbonic acid may lead to the formation of aqueous calcium bicarbonate leading to localized increases in porosity and permeability. Therefore, converting CH to CSH through the pozzolanic reaction by the addition of reactive silicas becomes a strategy to enhance the performance of cement admixtures in a CCS environment [Benge 2009].

Addition of epoxy-amine resin to Portland may enhance the resistance hydrated Portland cement to carbonation by reducing permeability and increasing chemical resistance. Permeability is reduced by replacing water in the system with a tightly crosslinked polymer network, consuming the pore space in the cement. Epoxy-amine systems are relatively inert to carbonic acid and may form microstructures within the set cement protecting the phases more susceptible to carbonation such as CH and CSH [Baldissera 2016].

Incorporation of advanced technologies to improve the resistance of the cement sheath also require lab testing to ensure mixability, appropriate rheology, thickening time, fluid loss and mechanical properties crucial to maintaining effective zonal isolation.

Cement Design considerations (Simulation sand Lab testing)

Primary cement was simulated using computational fluid dynamics and tailored cementing placement based on the ECD, Surface pumping pressure, down hole pressure profile, temperature and pressure for the fluid placement, casing stretch and stand-off. The design is fit for the purpose is endorsed by the 3D displacement efficiency of the fluids.

Inadequate pressure balancing during cementing operations can result in gas and fluid migration, leading to the formation of weak zones. Additionally, production depletion may adversely affect the integrity and performance of the cement.

To mitigate the issues and improve the success rate of cementing jobs, several measures can be undertaken:

- **Slurry Design:** Optimising cement slurry composition—including thickening time, rheology, and compressive strength—improves bonding and durability.
- **Hole Cleaning:** Thoroughly remove drilling fluids and cuttings to prevent contamination and ensure proper cement placement.
- **Centralization:** Centralizer casing for uniform cement distribution, minimizing channeling and enhancing cement integrity.
- **Wellbore Conditions:** Monitor temperature and pressure during cementing to prevent disruptions.
- **Cement + Resin System:** Use this system to reduce cement slurry carbonation.
- The implementation of advanced simulations, optimization of slurry properties, comprehensive laboratory testing, and meticulous operational planning have facilitated effective cement placement. Post-operation assessments demonstrate superior bonding and isolation, with robust zonal isolation and wellbore stability substantiated through computer modelling.

Cementing procedures adhered to a schedule established by temperature simulations. Additional evaluations related to the cementing process encompassed sedimentation analysis, SGSA, sensitivity, and compatibility studies. Specialized testing for slurry properties included CO₂-related analyses such as phenolphthalein testing, assessment of mechanical properties, permeability and porosity measurements, and electron microscope imaging; these procedures may extend over a period of up to three months. Incorporating resin into the slurry is anticipated to mitigate cement carbonation and further improve its mechanical characteristics.

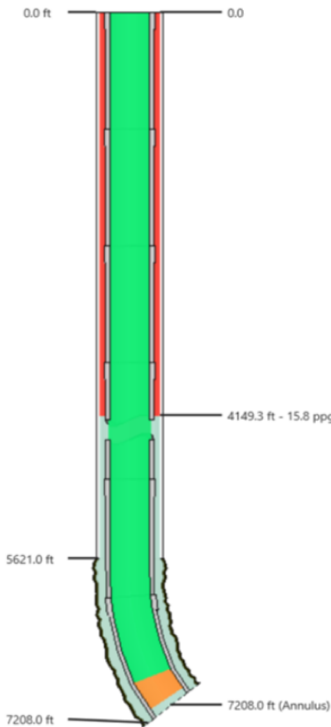


Figure 1—Well Schematic & Profile

Well profile

ECD

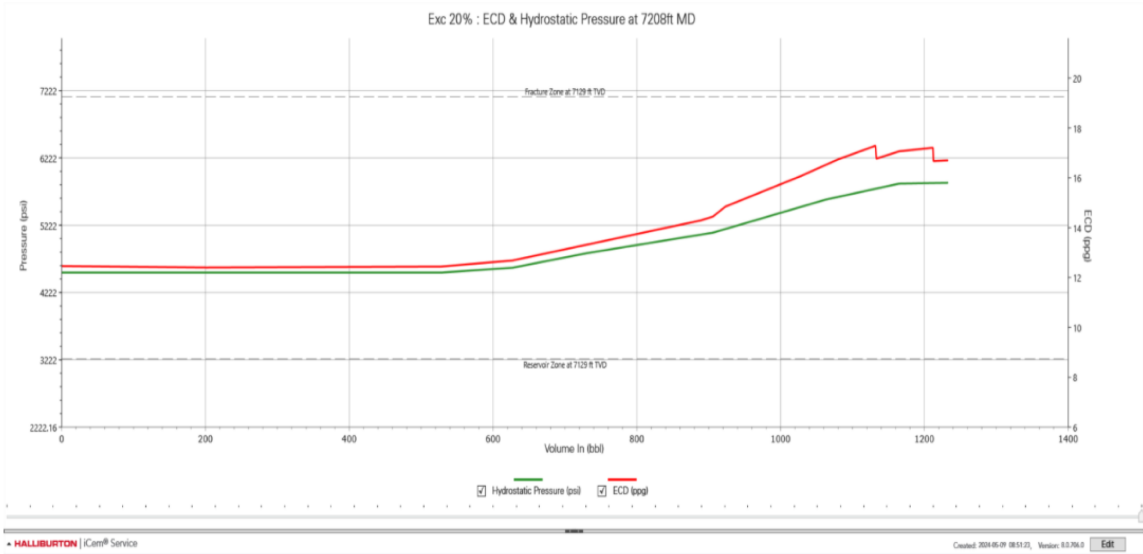


Figure 2—ECD Chart

Pumping Pressure

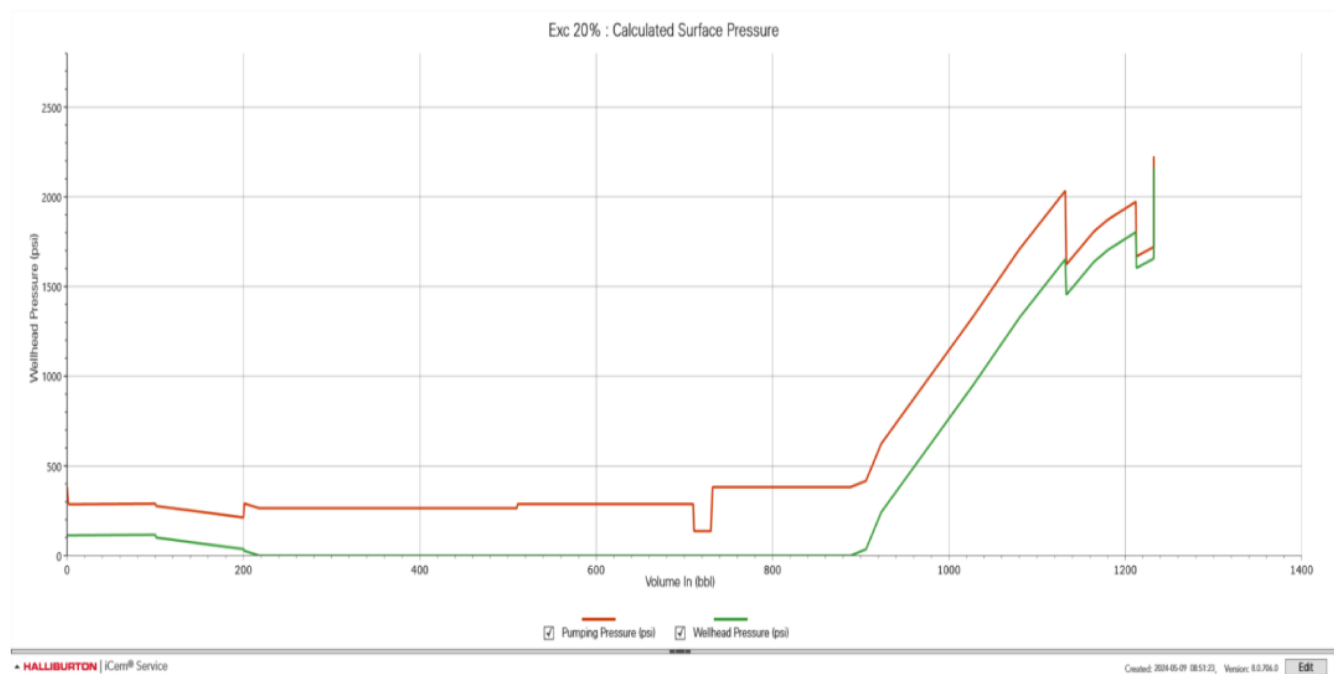


Figure 3—Surface Pressure Chart

Downhole Pressure Profile

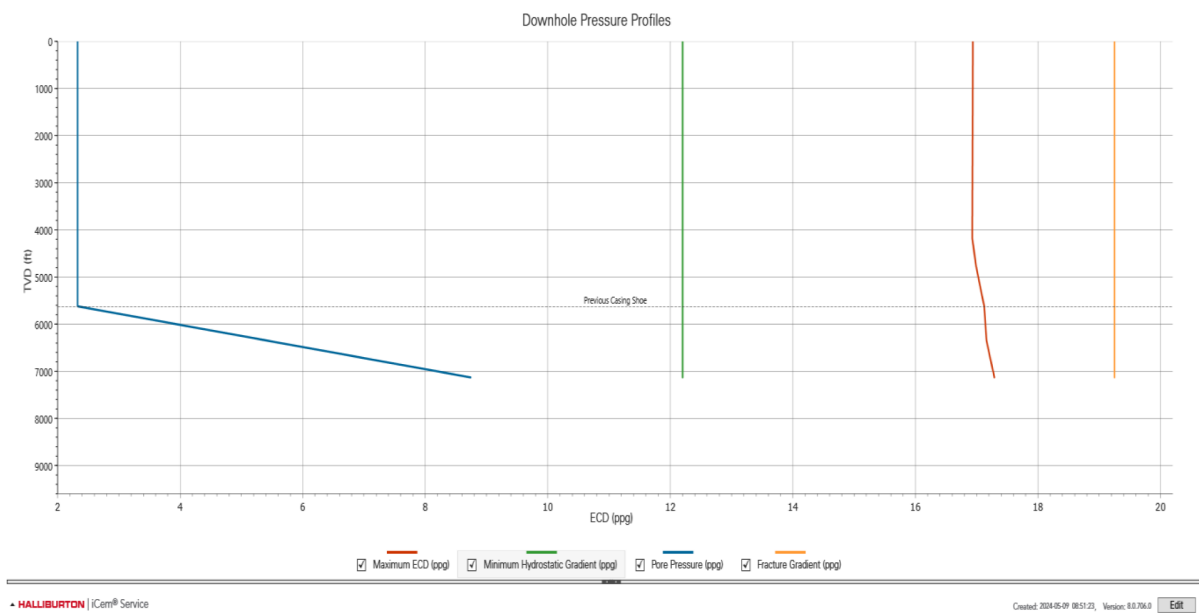


Figure 4—Downhole Pressure Profile

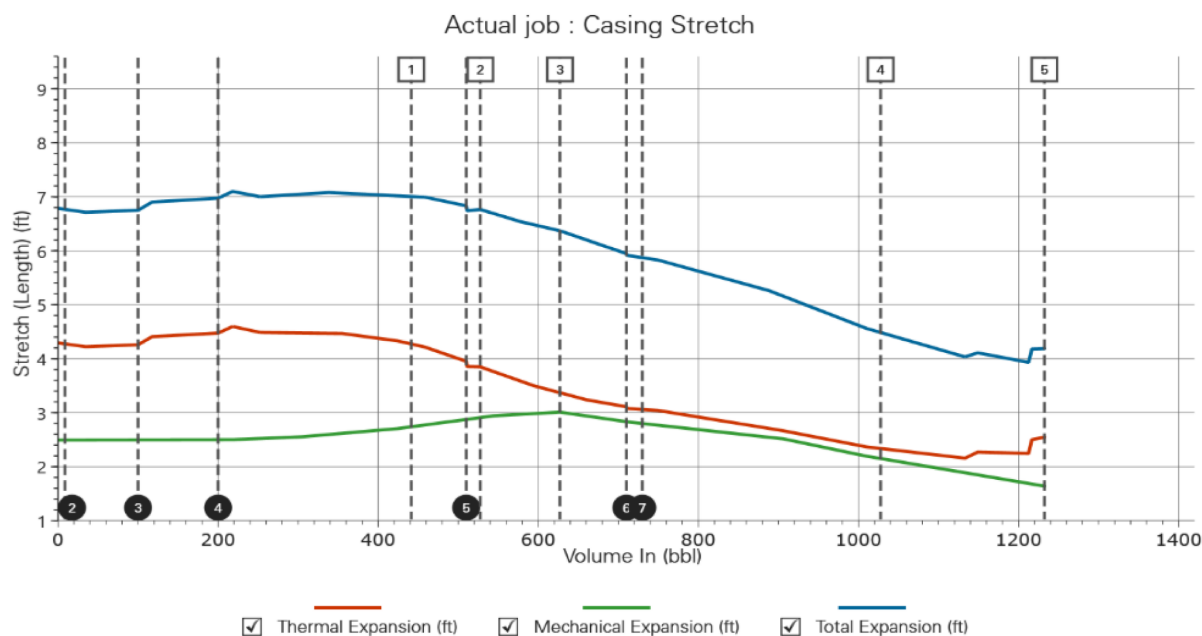


Figure 5—Casing Stretch

Contamination model

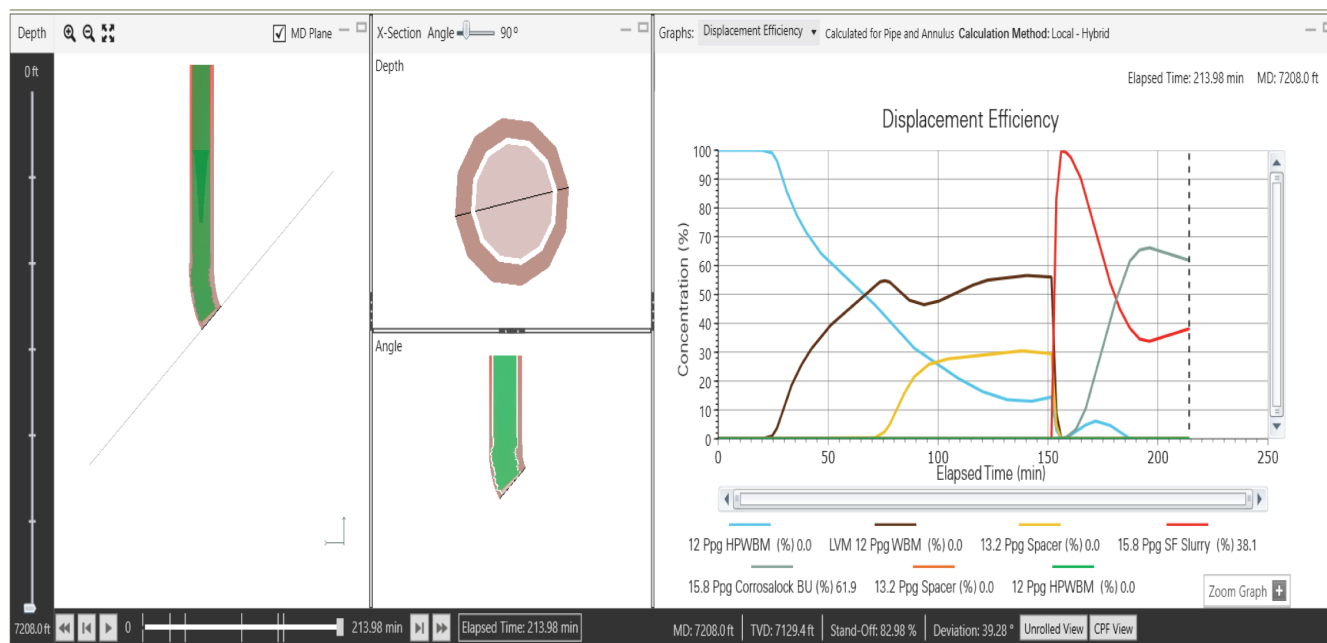


Figure 6—3D Simulation

Figure-6 is showing contamination simulation at downhole. It is expected to have 100% Cement at shoe without any contamination with mud and spacer.

Ultimate Compressive Strength

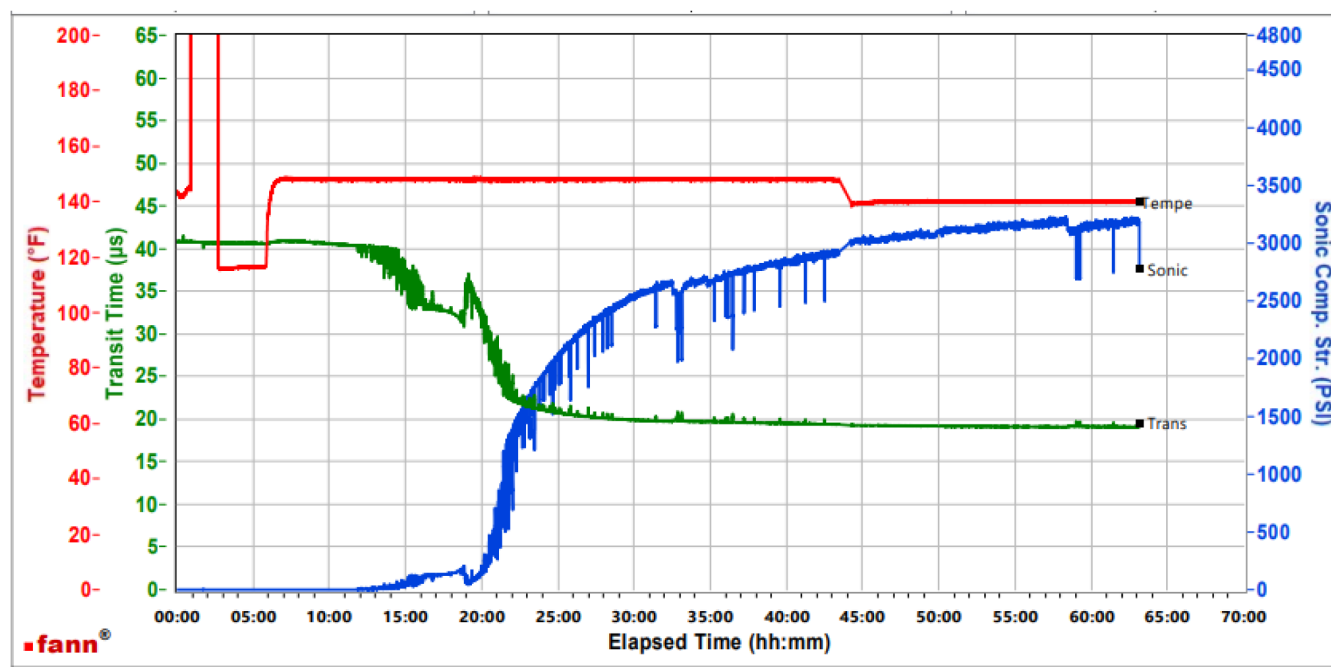


Figure 7—UCA Chart

Figure-7 is showing that in 20 hrs cement is reaching to 500 psi. Testing is done at BHCT to simulate the worst case scenario. It can be seen the strength continues to increase with time and reaches upto 3300 psi after 62 hrs.

Special Testing for the resin based system

The resin based cement slurry system is produced by separately mixing the cement slurry and epoxy resin, with hardener and other chemicals added according to well design specifications. A portion of the resin forms a film around the cement particles, creating a coating that assists with bonding. Additionally, the resin generates small spheres that fill the pore spaces within the composite matrix. This process reduces the effective porosity of the system and establishes an adhesive layer that shields the cement grains from CO₂-related degradation. The combination of the coating effect and matrix enhancement results in a composite system with lower permeability, which provides resistance to corrosion in both dry and aqueous supercritical CO₂ environments.

Phenolphthalein Testing

Special testing was performed for the resin-based cement slurry system. This innovative approach involves separately preparing the cement slurry and the epoxy resin. The hardener and other additives are incorporated as per the well design. During mixing, the resin partially coats the cement particles, creating a protective film that significantly enhances bond strength. Additionally, the resin forms microspheres that infiltrate the pore spaces within the cement matrix, effectively reducing porosity.

The dual action—coating and pore filling—results in markedly lower permeability compared to conventional systems. This not only minimizes fluid ingress but also imparts robust resistance to CO₂-induced corrosion. The adhesive resin layer shields cement grains from chemical attacks, particularly in wells exposed to dry or supercritical CO₂ environments, thereby extending the longevity and durability of the cement sheath.

Phenolphthalein testing is typically employed to assess the permeability and the degree of carbonation in the cement matrix. The method hinges on phenolphthalein's property of changing color in response to pH: a pink hue indicates uncarbonated, intact cement, while a colorless outcome signals carbonation or potential degradation. This allows for visual confirmation of the cement's integrity and the effectiveness of the resin coating in protecting against CO₂ ingress.

Regarding CO₂ testing, although no API (American Petroleum Institute) testing protocol is available or referenced for this specific resin-based system, in-house or tailored procedures are often used to simulate downhole exposure to supercritical CO₂. Important observations from such testing typically include measurements of changes in permeability, compressive strength, and signs of microstructural degradation over time. The expected outcome is a significant reduction in permeability, enhanced cement elasticity, and minimal to no evidence of CO₂-induced deterioration.

The combined effect of careful batch mixing, innovative resin system design, and comprehensive testing provides a cement sheath that is more elastic, less permeable, and robustly resistant to corrosion and chemical attack. This ensures wellbore integrity, especially in challenging environments where conventional cement systems may fail.

Phenolphthalein, a pH stain, provides visual cues of carbonation. Purple represents unaltered cement. The results of cement resin based cement slurry system (left) after one month of static supercritical CO₂ exposure at 100°F exhibit no noticeable CO₂ chemical alteration compared to conventional Portland cement (right).

Phenolphthalein testing leverages the pH-sensitive dye phenolphthalein to visually assess the carbonation and integrity of cement exposed to CO₂. When applied, phenolphthalein stains healthy, uncarbonated cement a distinctive purple or pink color, indicating high alkalinity and intact material. In contrast, sections that have undergone carbonation—a reaction that lowers the pH due to CO₂ ingress—remain colorless, signaling chemical alteration or potential degradation.

In the context of the resin-based cement slurry system, samples were subjected to one month of static exposure to supercritical CO₂ at 100°F. After this period, phenolphthalein was applied to both the resin-based and conventional Portland cement samples. The resin-based system retained the characteristic purple coloration, demonstrating its ability to resist chemical attack from CO₂ and maintain its structural integrity. By comparison, conventional Portland cement exhibited widespread colorless areas, evidence of significant CO₂-induced carbonation and deterioration.

These results highlight the superior performance of the resin-based cement: its protective coating and matrixfilling properties provide a robust barrier to CO₂, preventing the chemical reactions that typically degrade conventional cement in such environments. This enhanced resistance translates to greater longevity, durability, and reliability for wellbore cement sheaths in challenging CO₂-rich conditions. Refer to [Figure-8](#) below for results.

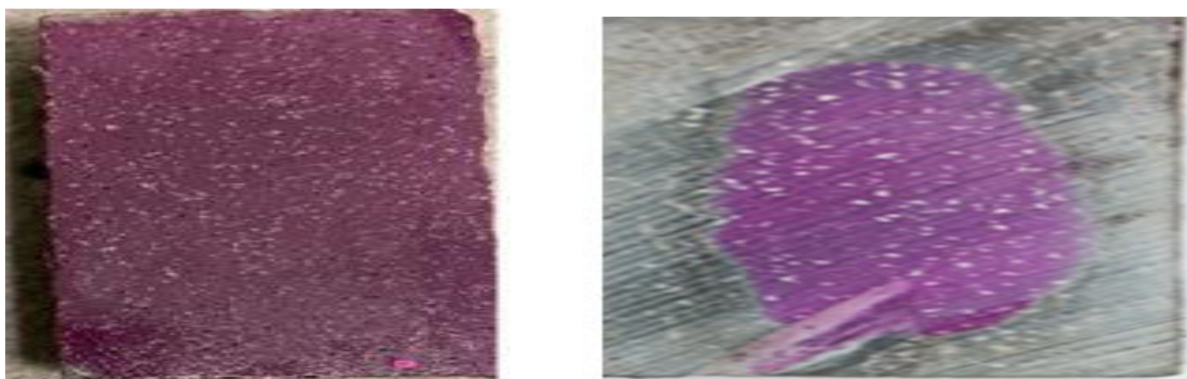


Figure 8—Phenolphthalein Testing

CO₂ Testing Methodology

Due to the absence of a standardized API protocol for this unique resin-based cement system, customized laboratory procedures are implemented to simulate exposure to supercritical CO₂ under downhole conditions. The process begins by preparing the cement slurry, which is then cured in 1-inch diameter molds under designated pressure and temperature parameters. Once cured, the cylinders (approximately 3 inches in length) are trimmed perpendicular to their axis, resulting in smooth samples between 2 and 2.5 inches long.

For testing, the samples are placed on a perforated platform within corrosion-resistant autoclaves, as illustrated in Figure 9. Two primary CO₂ exposure environments are utilized:

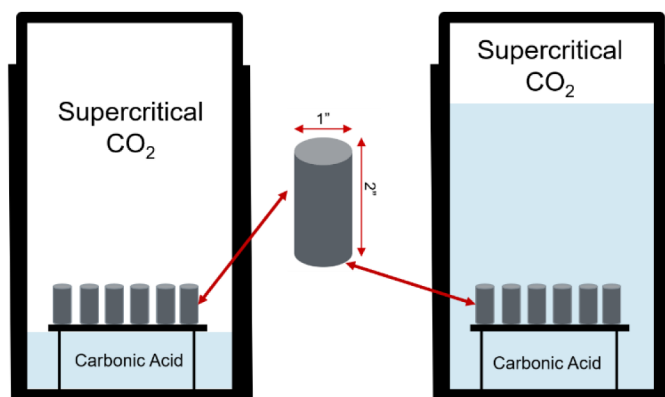


Figure 9—Phenolphthalein Testing

- **Carbonic Acid Environment:** The cement cylinders are fully submerged in in-situ carbonic acid, generated by introducing 800 ml of deionized water and 99% pure CO₂ into the chamber at elevated pressure.
- **Supercritical CO₂ (SC-CO₂) Environment:** Only 90 ml of deionized water is added, and the cylinders are positioned above the water line, exposing them to a CO₂-saturated water vapor atmosphere.

After loading the samples, the autoclave lid is sealed. The chamber is purged with CO₂ gas, venting through the thermocouple connection to ensure saturation. Thermocouples are installed both in the chamber lid and heating jacket, with connections to a temperature controller for precise regulation.

Initial pressure is set just high enough to prevent water from boiling at the test temperature, but low enough to accommodate thermal expansion without risking overpressure (refer to Figure 3: Saturation Pressure Curve for Water from Wagner/Pruss 1993). The system is then pressurized by connecting the autoclave to a CO₂ cylinder, with pressure regulated and monitored using the cylinder's gauges. It is crucial to verify that the CO₂ source has sufficient volume for the duration of the test; if depletion occurs, the autoclave inlet valve should be closed and the cylinder replaced before repressurizing.

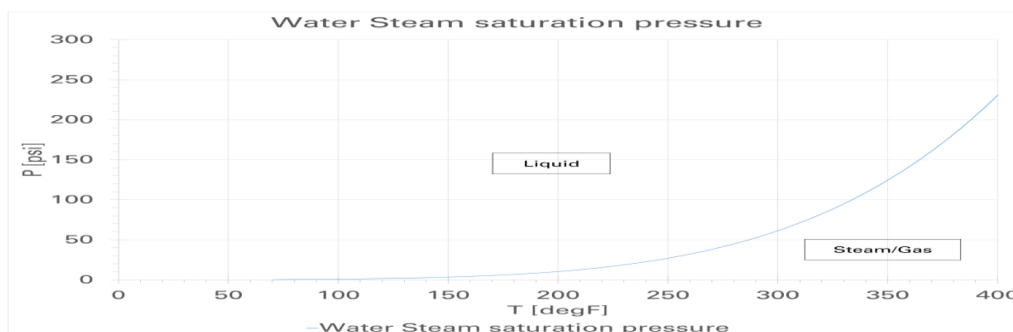


Figure 10—Saturation Pressure Curve for Water (Wagner/Pruss 1993)

Once the target pressure is achieved, the autoclave inlet valve is closed and the feed hose is safely depressurized. Temperature is programmed into the controller, and the heater is activated. During heat-up, control is managed by the jacket thermocouple for faster feedback; once the desired temperature is reached, the system switches to the chamber thermocouple to ensure chamber conditions remain steady.

Throughout the test, regular inspections are essential to confirm that temperature and pressure remain stable. Repressurization may be necessary if pressure drops, and CO₂ cylinders should be replaced as needed to maintain consistent supply. When the test concludes, the heater is turned off but temperature monitoring continues. Pressure release is performed only after the temperature falls below 180°F, preventing flash evaporation of water. The gas supply from the CO₂ cylinder is shut off, and pressure is gradually released via the syringe pump or a controlled outlet. After pressure is fully relieved, the autoclave is carefully opened, samples are photographed in situ, and cylinders are extracted for further analysis. Water from the chamber is collected for subsequent analytical testing.

Assessing Depth of CO₂ Penetration

To measure the extent of carbonation, one exposed cylinder is cut lengthwise. Phenolphthalein dye is applied to the fresh surface—uncarbonated, high-alkalinity zones will turn pink, while carbonated, chemically altered regions will remain colorless (a method primarily applicable to Portland-based systems). This visual comparison is then benchmarked against an unexposed control sample to quantify the depth of CO₂ ingress and evaluate the protective performance of the resin-based cement system.

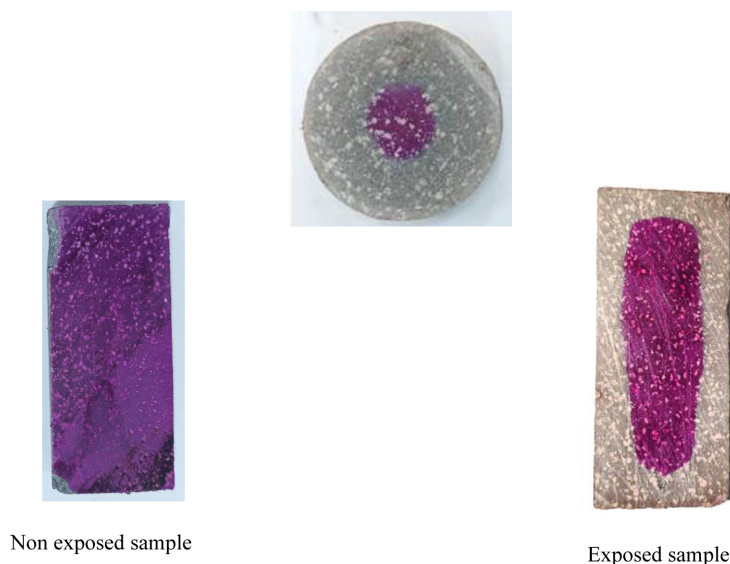


Figure 11—Assessing Depth of CO₂ Penetration

CO₂ Corrosion Test – Setup

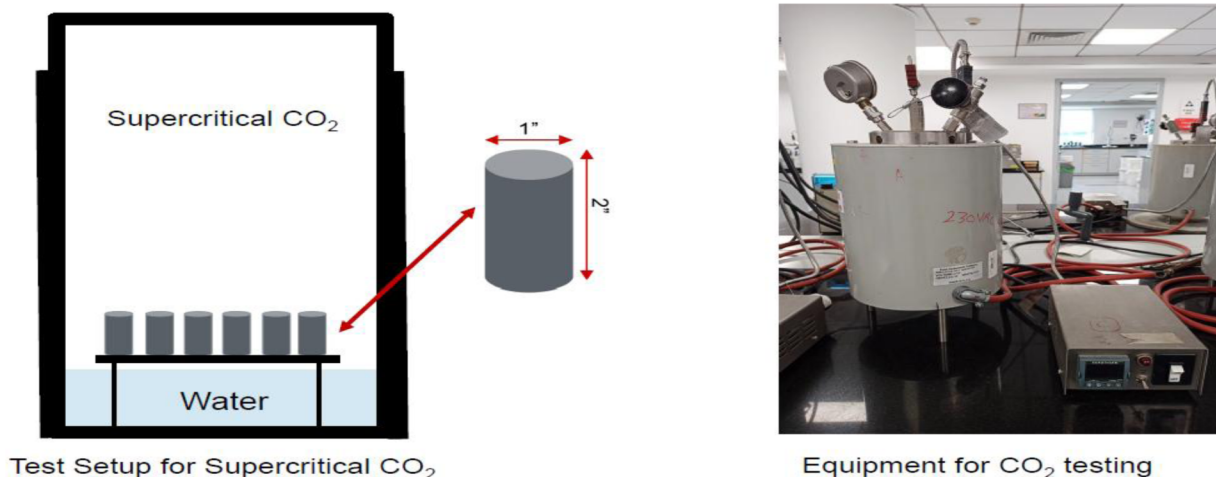


Figure 12—Corrosion Test Setup

Results obtained

Shear Bond Strength.

- Resin-Cement System: 1614lbf.
- Conventional Cement: 945lbf. The resin-cement system showed a 70% increase in shear bond strength, indicating a much stronger bond between casing and cement.

The resin-cement system demonstrated a 70% increase in shear bond strength, indicating a significantly stronger bond between the casing and the cement.

Mechanical Properties

Table 2 is showing Lower Young's modulus in resin-cement means greater flexibility, which helps resist stress from pressure and temperature cycling

Table 2—Mechanical Properties

Property	Conventional Cement	Resin-Cement System
Static Young's Modulus	2.33 million psi	1.40 million psi
Dynamic Young's Modulus	3.46 million psi	0.98 million psi
Static Poisson's Ratio	0.195	0.18
Dynamic Poisson's Ratio	0.274	0.23

Table 3—Test Conditions

Test Condition	Pre Cured samples	Super critical CO2 @220f/2000 psi		
Duration	07 days	30 days	90 days	180 days
Porosity (%)	23.12	4.779	9.988	6.307
Permeability (mD)	.035	.036	.0628	.045

Compressive Strength.

- UCA Test: 500 psi after 12-36 hours
- Crush Test = 2200-2500 psi after 24hours. Resin-cement develops strength more slowly in UCA but achieves high compressive strength in crush tests, indicating robust long-term performance.

Thickening Time.

- Target: ~06 hours
- Results: 06:15 to 08:15 hours across pilot and confirmation tests. Resin-cement slurry provided ample pump time and consistent thickening behaviour, even with batch-mix simulation.

Sedimentation Stability (BP Settling).

- Top-to-bottom density variation: <2% (acceptable range)The slurry showed minimal segregation, ensuring uniform placement.

Field Performance.

- 08 wells cemented with resin-cement showed:
 - No sustained casing pressure (SCP)
 - Good zonal isolation
 - No post-job remedial work required Field validation confirms resin-cement's reliability under extreme conditions

The resin-cement system outperforms conventional cement in:

- Bond strength
- Flexibility
- Resistance to stress cycling
- Long-term sealing integrity

It is especially suitable for:

- SWAG/CCS wells
- CO₂ injection environments
- Wells with high risk of micro-annulus or SCP

Job execution and details

After comprehensive engineering and laboratory testing, a detailed cement program was developed to outline the execution of the cement job. For the 9 5/8 casing cementation, the cement slurry was designed with both lead and tail slurries; the lead component consisted of conventional cement to address cost considerations, while the tail slurry comprised a cement and resin system. Typical volumes for the tail slurry ranged from 200 to 250 barrels, and the batch mixing and pumping process involved preparing specified volumes of resin and cement slurry at a density of 15.80 lb/gal. During the primary cement job, 200 barrels were mixed and pumped, followed by a plug bump test at 2500 psi. The test confirmed that the Non-Return

Valve (NRV) held successfully, with approximately 35 barrels returned to the surface, indicating effective displacement and integrity of the cement barrier.

For the 7-inch liner job, two separate batches of cement and resin (lead and tail, respectively) were batch mixed and pumped. Lead densities of approximately 15.8 ppg and tail densities of approximately 16.7 ppg were used. The cementing operation matched the anticipated pressure and density profiles based on simulations. Cement returns at the surface exceeded 25 bbls.

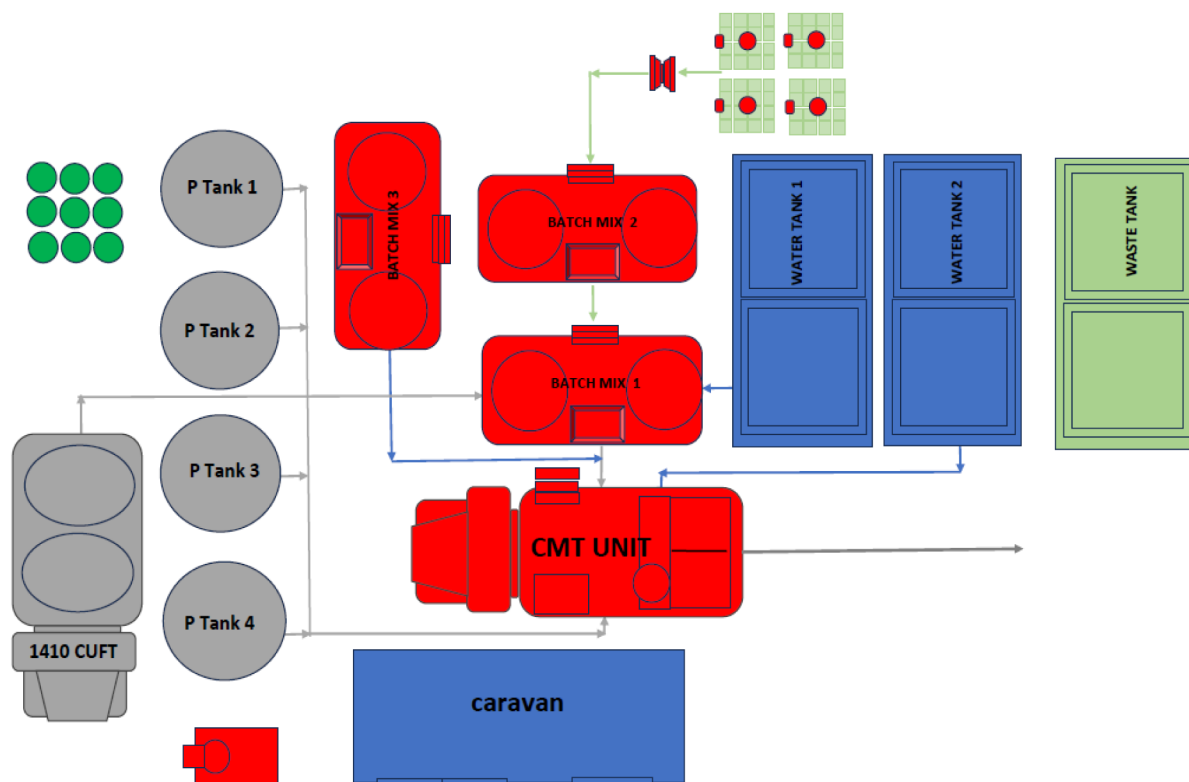


Figure 13—Equipment Layout

Finally, job execution details and cement evaluation, including monitoring volumes pumped and returned, pressure testing, and post-job analysis, provide confirmation of proper placement and the anticipated performance enhancements. The sequence of operations, duration, and detailed observations would be documented to support conclusions and continuous improvement in future applications.

Results, Observation & Conclusion (Impact of the Cement and Resin Slurry System on Sustainability Initiatives)

A. Advancing Environmental Responsibility in Well Construction

The adoption of the innovative cement and resin slurry system, as outlined in the operational overview, marks a significant advancement in sustainable well construction practices. This approach, which strategically combines conventional and resin-enhanced cement slurries for both casing and liner jobs, aligns closely with contemporary goals for minimizing environmental impact and enhancing long-term well integrity.

B. Reduction in Environmental Footprint

By optimizing cement volumes and densities—such as the use of tailored lead and tail slurries—the system reduces overall material consumption. The ability to precisely batch mix and pump only the required amounts curtails waste generation at the wellsite. Additionally, the improved displacement efficiency, evidenced by high cement returns at surface and minimal excess, further

limits unnecessary usage of resources, thus decreasing the carbon footprint associated with cement production, transportation, and disposal.

C. Enhanced Well Integrity and Longevity

A core pillar of sustainability in oil and gas operations is the assurance of long-term well integrity, reducing the risk of leaks and associated environmental hazards. The superior bonding and sealing properties offered by the cement-resin tail slurry provide robust isolation, particularly in challenging environments prone to micro-annulus or sustained casing pressure (SCP). This reliable barrier mitigates risks of fugitive emissions—such as methane or CO₂ migration—into surrounding formations or the atmosphere, supporting broader emission reduction commitments.

Additionally, the formulation's resistance to carbonation—a process where CO₂ reacts with the cement matrix, potentially compromising its structural integrity—is a significant advantage. By incorporating resin into the cement system, the slurry enhances durability against CO₂-induced deterioration, preserving the long-term sealing capacity of the well. This added defense not only maintains zonal isolation but also bolsters confidence in carbon capture, utilization, and storage (CCUS) applications, ensuring that injected CO₂ remains securely contained over time.

D. Adaptability to CO₂ and Challenging Environments

The system's suitability for CO₂ injection wells and applications with high annular pressure risk demonstrates its forward-thinking design for carbon capture and storage (CCS) projects. Effective zonal isolation in such contexts ensures that injected CO₂ remains securely contained, enhancing the viability and safety of CCS as a climate mitigation strategy. By supporting these next-generation energy solutions, the slurry system contributes to the transition toward lower-carbon operations.

E. Operational Efficiency and Resource Optimization

The meticulous engineering and laboratory validation that underpin the slurry design contribute to operational excellence. Fewer remedial jobs and reduced non-productive time (NPT) mean less fuel usage and lower emissions from heavy equipment. The successful primary cementations, as indicated by effective plug bump tests and cement returns, minimize the likelihood of repeated interventions, further lowering environmental disturbance and enhancing cost efficiency.

Conclusion

In summary, the deployment of the cement and resin slurry system exemplifies a holistic approach to sustainability in well construction—optimizing resource use, ensuring durable environmental barriers, and supporting technologies critical to a lower-carbon future. Notably, the system's inherent resistance to carbonation a process where CO₂ reacts with and can progressively weaken traditional cement matrices marks a significant advancement. By integrating resin, the slurry maintains structural integrity and prolongs sealing capacity even in CO₂-rich environments, safeguarding against the long-term deterioration that could compromise zonal isolation. This methodology not only safeguards the immediate well environment but also aligns with industry-wide efforts to implement more responsible, resilient, and sustainable practices across the full lifecycle of energy development, particularly in applications such as carbon capture and storage where carbonation resistance is essential.

Nomenclature

SCD	= Smart casing design
HCD	= Heavy Casing Design
CWD	= Casing While Drilling
MPWD	= Managed pressure While Drilling
ALWTC	= Advance Light Weight thixotropic cement
BHA	= Bottom Hole Assembly

CBL = Cement Bond Log
ECD = Equivalent Circulating Density
LCM = Loss Circulation Material
ERD = Extended Reach Drilling
TT = Thickening Time
LWB = Light Weight Blend
PPG = Pounds Per Gallon
PSI = Pound Per Square Inch
SVF = Solid Volume Fraction
UAE = United Arab Emirates
UCA = Ultra Sonic Cement Analyzer
USD = United States Dollar
VDL = Variable Density Log
CCU = Carbon Capture Sequestration
SWAG = Simultaneous Water and Gas.
CCUS = Carbon Capture, Utilization, and Storage (CCUS)

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